

CMP-10 · COMPETENCY FRAMEWORKS

Assessing competence — evidence, rubrics and moderation

How a competency claim is tested, why rubrics fail, and what
moderation is actually diagnosing

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AI proposes. The professional disposes.

— Summary

An assessment is an inference from a sample of evidence to a claim about capability, and it can fail in two directions with very different costs. This document sets out the evidence types and what each can and cannot show, how to write a rubric descriptor that two assessors read the same way, the difference between compensatory and conjunctive scoring and when each is legitimate, how moderation works and what its disagreements actually diagnose, and why self-assessment on its own carries no assessment weight.

— About this document

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Assessing competence — evidence, rubrics and moderation

In one paragraph. An assessment is an inference: from a sample of evidence, to a claim about what a person can do in situations nobody has observed. It can fail in two directions with very different costs. This document sets out the evidence types and what each can and cannot show, how to write a rubric descriptor that two assessors read the same way, the difference between compensatory and conjunctive scoring and when each is legitimate, how moderation works and what its disagreements actually diagnose, and why self-assessment on its own carries no assessment weight — together with the three things it is genuinely good for.

Who this is for. Anyone who assesses competence and has to defend the decision: certification assessors, heads of function running internal competency schemes, and employers whose promotion and deployment decisions rest on a rating someone produced.

— 1. What an assessment is claiming, and how it fails

Every competence decision has the same shape. A person's capability across an entire working life is unobservable. A sample of it is observed — some artefacts, an examination, a conversation — and from that sample an inference is drawn about behaviour in situations that were not sampled.

Two properties determine whether the inference holds.

Validity — is the evidence about the thing being claimed? A written examination on forecasting methods is valid evidence of knowledge of forecasting methods. It is weak evidence of whether the candidate will hold a forecast under commercial pressure, because that situation is not in the sample. Most assessment failures are validity failures wearing the clothes of rigour: the instrument is precise, consistent and measuring something adjacent to the claim.

Reliability — would the same evidence produce the same decision with a different assessor, on a different day? An unreliable process is unfair in a way that is invisible to everyone inside it, because each assessor believes their own judgement to be the standard.

The two failure directions have asymmetric costs, and the asymmetry should be decided deliberately rather than absorbed by default:

- A **false pass** puts an unqualified person into a role where the harm is borne by third parties — a project, a client, a lender, a workforce — and it devalues the credential for everyone who holds it.
- A **false fail** harms one identified individual, who is usually in front of you, and who will appeal.

Because the false fail is visible and the false pass is not, assessment systems drift toward passing. The counterweight is not severity; it is written standards, evidence rules and moderation.

— 2. Evidence types, and what each can carry

The hierarchy is set out in CMP-02 §4. Here is what each type does under examination, and how each is gamed.

Evidence type	Carries	Cannot carry	Typical failure	Control
Work product authored by the candidate	Method choice, judgement in real conditions, standard of output	Whether the conditions were hard; sole authorship	Team output presented as personal	Require the reasoning to be visible, and probe it in discussion
Work product contributed to	Capability within a defined scope	Ownership of decisions made by others	Contribution overstated in retrospect	Require identifiable contribution recorded at the time
Structured professional discussion	Depth behind the artefact; reasoning under questioning	Anything the questioner did not probe	Interview skill rewarded over competence	Question from the candidate's own artefacts only; use a common question frame
Direct observation	Behaviour and interaction, which artefacts never capture	Anything outside the observed window	Performance for the observer	Observe routine work, not set-piece events
Attestation by a named referee	Corroboration of scope, duration and standing	Detail; independence from the referee's judgement	Reciprocal or reputational attestation	Require the referee's relationship, period and specific observation
Knowledge examination	Applied knowledge and reasoning against a common standard, reliably and at scale	Behaviour under real pressure	Treated as sufficient on its own	Combine with evidence types above
Self-report	A map of where to look	Capability	Uniformly favourable	Use for routing only — see §6

Two rules that resolve most disputes. **Where types conflict, the higher-ranked type wins** unless there is a stated reason to prefer the lower. And **absence of evidence is not evidence of absence** — it is absence of evidence, and the assessment record should say "not evidenced" rather than assigning a low level. The distinction matters at appeal, and it is the single most common recording error.

— 3. Writing a rubric that two people read the same way

Most rubrics fail in one of four ways, and all four are visible on the page before anyone is assessed.

Adjectival ladders. "Basic / good / strong / excellent understanding." Nothing is described; the assessor supplies the meaning, and two assessors supply different meanings.

Global impression. One overall rating for a whole competency area. It correlates with how articulate the candidate was, and it cannot be defended clause by clause at appeal.

Double-barrelled criteria. "Selects an appropriate forecasting method **and** communicates it to stakeholders." A candidate who does the first well and the second badly cannot be scored, and different assessors resolve the conflict differently.

Unbounded top levels. "Exceptional insight." Nobody can say what would satisfy it, so it is awarded on impression or never awarded at all.

The construction that works has three parts in every descriptor: **an observable action, a condition, and a standard.**

Action: selects a forecasting method — condition: where procedure does not prescribe one — standard: and records the method, the alternatives rejected and the assumptions the answer depends on.

Four supporting rules:

1. **One criterion, one judgement.** Split anything containing "and" that could be met by halves.
2. **Every level is bounded above and below.** An assessor must be able to say why this is a 3 and not a 4.
3. **Include an insufficient-evidence outcome distinct from the lowest level.** Otherwise "nothing submitted" is scored as "foundation", which is a different claim.
4. **Write the descriptors so that the difference between adjacent levels is a difference in the work, not in the adverb.** If levels 2 and 3 differ only by "consistently", the rubric will not moderate.

3.1 A worked rubric: Forecasting and EAC (PCL-AI competency 5)

Estimate at completion (EAC). Assessed from candidate-authored artefacts and a structured discussion of them.

Level	Descriptor	Evidence that satisfies it
0 — Insufficient evidence	No artefact authored or materially contributed to by the candidate, or the contribution cannot be identified.	—
1 — Foundation	Computes a forecast using the method prescribed by procedure, and identifies inputs that appear inconsistent with the project position.	A completed forecast to procedure; a query raised on an input that was subsequently corrected.
2 — Practitioner	Produces the periodic forecast for an owned scope, selecting from the methods the procedure permits, and explains the movement since the previous period in terms of its drivers.	Consecutive periodic forecasts with movement analysis; a forecast relied on without correction by the next reviewer.
3 — Professional	Selects the forecast method where procedure does not prescribe one; records the method, the alternatives rejected and the assumptions on which the answer depends; states a range where a range is warranted; and maintains the forecast under challenge, changing it on evidence and not on pressure.	A dated method-selection note; a forecast revised in a period when no external event forced it, with the basis; a documented challenge and its resolution.
4 — Leader	Sets the forecasting standard applied by others across projects — method rules, review cadence, treatment of undisclosed exposure — and can demonstrate the standard operating, including a case where it produced an unwelcome answer that was nonetheless reported.	An issued forecasting standard in use, with the candidate's authorship; back-test evidence across projects; a documented escalation route used at least once.

Note what is doing the work: level 3 is distinguished from level 2 by **who chose the method and whether the choice is recorded**, and level 4 from level 3 by **whether others apply the rule**. Neither distinction requires an assessor's opinion about quality.

— 4. Scoring: compensatory, conjunctive, and the choice between them

Compensatory scoring allows strength in one criterion to offset weakness in another — typically an average across criteria against a threshold. **Conjunctive** scoring requires a minimum on each criterion independently; no amount of excellence elsewhere compensates.

Compensatory scoring is appropriate where the competencies are genuinely substitutable in the role. Conjunctive scoring is required where a specific weakness is unacceptable regardless of other strengths — and in this framework at least two areas qualify: **responsible AI** and **human validation or human accountability**. A person who is outstanding at analysis and does not validate output is not a net positive, and an average conceals exactly that.

The practical construction most schemes should use is **conjunctive on a named subset, compensatory on the remainder**, with the subset published in advance. §8 shows how a compensatory average conceals a profile that no assessor would accept if they saw it laid out.

How PCI's own examination scores, stated plainly. It is criterion-referenced, not graded on a curve: candidates are measured against a standard, not against each other, so a strong cohort does not raise the bar and a weak one does not lower it. The examination is 90 minutes, four-option multiple choice with a single best answer, written across recall, application and analysis and weighted toward application, and results are returned immediately with a domain breakdown. The pass mark configured in the platform is **65 %**, and the definitive standard will be set by a **modified-Angoff standard-setting study** — a method in which subject-matter experts judge, item by item, the probability that a minimally competent candidate answers correctly, and the aggregate of those judgements sets the mark. **The number of items is not published, because it has not been decided**; it will be confirmed by a formal job-task analysis. EXB-05 and EXB-06 own these two studies.

— 5. Moderation: what it is for and what it finds

Moderation is not a second opinion. It is a control on the assessment process, and its outputs diagnose the instrument at least as often as the candidate.

The mechanics that make it work:

- **Independent first.** Two assessors score without seeing each other's marks. A "review" of an existing mark anchors to it and detects almost nothing.
- **A defined sample.** Full double-marking is rarely affordable. A stated proportion is moderated, plus every decision near a threshold, every fail, and every appeal [\[CONFIRM: moderation sampling rate for PCI competence assessments\]](#).
- **Reconciliation by evidence, not by rank.** Disagreements are resolved by both assessors returning to the artefact and to the rubric. Where they cannot agree, a moderator or chair rules, and the ruling is recorded with its reasoning.
- **Disagreement is analysed, not just resolved.** Every disagreement has one of four causes: the rubric is ambiguous, an assessor is miscalibrated, the portfolio is disorganised so the two assessors read different evidence, or the case is genuinely borderline. Only the last is about the candidate, and only the last requires no corrective action.

- **Calibration before the cycle.** Assessors score the same anonymised sample and discuss divergence before assessing anyone real [\[CONFIRM: assessor qualification and calibration requirements\]](#).
- **Conflict of interest declared and acted on.** An assessor does not assess a person they manage, have trained, have been employed alongside recently, or have a commercial relationship with. Separation of training from assessment is a founding principle of personnel certification and is not a formality — it is what makes the decision worth anything to a third party.
- **Records retained.** The evidence considered, the rubric applied, the levels awarded, the disagreements and their resolution. An assessment that cannot be reconstructed cannot be appealed fairly, and CER-07 — Results, scoring, appeals and complaints depends on it.

— 6. Why self-assessment alone carries no weight

This is not a claim about honesty. It follows from three structural features of the act, none of which improves with better intentions:

There is no external referent. The rater compares their work to their own experience of the work. A person whose entire exposure to forecasting has been one organisation's method rates themselves against that method, and cannot rate themselves against practice they have never seen.

The rater has an interest in the outcome. Where a self-assessment feeds a promotion, a fee or a credential, the rating is a submission in a process, and submissions are optimised.

The specific failure is invisible from inside. To know that your method selection is inadequate you must know what an adequate one looks like. A person who has never seen a method-selection note recorded does not experience its absence as a gap; they experience their own practice as normal. This is why self-assessment is least reliable exactly where it would be most useful — at the boundary between practitioner and professional.

Self-assessment nevertheless has three genuine uses, and a scheme that discards it loses them:

1. **Routing.** It tells an assessor where to look and lets a candidate assemble evidence efficiently.
2. **Structuring a professional discussion.** "You rated yourself professional here — show me the artefact" is a productive opening, and the answer is informative whichever way it goes.
3. **The gap between self and manager rating.** The disagreement is the signal. A large gap in either direction is worth an hour of conversation, and the direction tells you which hour it is: a person rating themselves well above their manager may be misreading the standard, and a person rating themselves well below may be evidencing something that is never seen.

Used for those three purposes, self-assessment is valuable. Used as a rating that stands on its own, it produces a number that reflects confidence and role identity, and it should not be recorded as an assessment outcome.

— 7. How this goes wrong

The rubric is written after the assessments. Standards are then reverse-engineered from decisions already made, which is unappealable and, when it surfaces, indefensible.

"Not evidenced" is recorded as "foundation". Two different findings. One says the candidate cannot do it; the other says nobody looked, or the candidate did not submit. Conflating them produces both unfair fails and unjustified passes, and it destroys the diagnostic value of the record.

Moderation is a signature. A moderator who reviews rather than independently scores anchors to the first mark. Agreement rates then look excellent and mean nothing.

Disagreement is resolved by seniority. The more senior assessor's view prevails without returning to the evidence. The scheme has now moved from criterion-referenced to person-referenced, and the analysis in §5 that would have found the rubric defect never happens.

A compensatory average hides an unacceptable profile. See §8. Averages are convenient and they conceal precisely the pattern a conjunctive rule exists to catch.

Assessors are never calibrated. Each is internally consistent and they differ from one another by half a level, which is invisible until a candidate assessed by the strict one appeals and cites a colleague assessed by the lenient one.

Interview performance is scored as competence. The structured discussion probes artefacts. When it becomes a general conversation, it measures fluency, which the assessment does not claim to certify and which correlates with things a fair process must not select for.

The assessment cannot be reconstructed. No record of what evidence was considered against which criterion. The first appeal exposes it, and the scheme's credibility does not recover quickly.

The examination is treated as the whole assessment. A knowledge examination is reliable, scalable and valid for what it samples. It does not sample behaviour under pressure. Scheme designers who forget this end up certifying knowledge and describing it as competence.

— 8. Worked example: a moderation reconciliation

Illustrative figures. Two assessors independently rate one candidate against five competencies on the four-level scale, coded 1 = foundation, 2 = practitioner, 3 = professional, 4 = leader.

#	Competency	Assessor A	Assessor B	Difference (A – B)
1	Cost management	3	3	0
2	Forecasting and EAC	3	2	+1
3	Predictive analytics	2	2	0
4	Project controls governance	4	3	+1
5	Human validation	2	3	–1

Agreement measures

Exact agreement: 2 criteria of 5 = 40 %. Adjacent agreement, meaning within one level: 5 of 5 = 100 %. Mean absolute difference: $(0 + 1 + 0 + 1 + 1) \div 5 = 3 \div 5 = 0.6$ levels. Mean signed difference: $(0 + 1 + 0 + 1 - 1) \div 5 = 1 \div 5 = +0.2$ levels — assessor A is running two-tenths of a level higher than assessor B on this candidate, which is a calibration observation and not yet a finding, because one candidate is not a sample.

Reconciliation, and what each disagreement diagnosed

- Criterion 2. Assessor A cited a dated method-selection note; assessor B had not seen it, because it sat in an appendix not referenced from the competency claim. Agreed at **3**. Cause: **portfolio organisation**. Action: require an evidence index mapping each artefact to the competencies it supports.
- Criterion 4. Assessor A treated authorship of a governance procedure as leader level. Assessor B applied the rubric, which requires the rule to be in use by others and a failure to have been owned; neither was evidenced. Agreed at **3**. Cause: **assessor calibration**. Action: calibration note issued to all assessors with this rubric row as the example.
- Criterion 5. Assessor B credited a validation record that assessor A read as a bare sign-off. On reading it together, the record named a date and a reviewer but not what was checked against what. Agreed at **2**. Cause: **rubric ambiguity** — the descriptor said "records the validation" without stating the minimum content. Action: rubric amended to require the independent source, the coverage and the finding.

Agreed profile: **3, 3, 2, 3, 2**.

Why the scoring rule matters. Suppose the decision rule is conjunctive on criteria 1, 2 and 4 at professional (3) or above, and practitioner (2) or above on the rest. The agreed profile meets it.

Under a compensatory rule instead — a mean of at least 2.6 — the same profile passes: $(3 + 3 + 2 + 3 + 2) \div 5 = 13 \div 5 = 2.6$. But so does the profile 4, 4, 1, 3, 1: $(4 + 4 + 1 + 3 + 1) \div 5 = 13 \div 5 = 2.6$ — a candidate at foundation level in predictive analytics **and** in human validation, passing on the strength of two leader-level ratings elsewhere. That is the profile a conjunctive rule exists to stop, and the average made it invisible.

The finding that matters. Of three disagreements, one was about the candidate and two were about the assessment system. That ratio is normal in a young scheme, and it is the reason moderation output is reviewed by whoever owns the rubric — not filed with the candidate's result.

— 9. Checklist

Before running an assessment, and before defending one:

- [] Is every criterion written as an observable action, a condition and a standard?
- [] Does any criterion contain an "and" that could be half met?
- [] Is each level bounded above and below, so an assessor can say why not the next one up?
- [] Is there an insufficient-evidence outcome distinct from the lowest level, and is it being used?
- [] Which criteria are conjunctive, and was that published before the assessment?
- [] Do the evidence types actually sample the claim, or only the knowledge adjacent to it?
- [] Did two assessors score independently before comparing?
- [] Are moderation disagreements classified by cause — rubric, calibration, portfolio, genuine borderline?
- [] Does the moderation output go to whoever owns the rubric?
- [] Were assessors calibrated on a common sample this cycle?
- [] Are conflicts of interest declared, and is anyone assessing someone they trained or manage?
- [] Could a third party reconstruct this decision from the record in twelve months?
- [] Is any self-assessment being recorded as an outcome rather than used for routing?

— Related

- [CMP-01 – The PCI competency framework – overview](#) — what is being assessed and how each competency is written
- [CMP-02 – Competency levels and how they are evidenced](#) — the level scale and the evidence hierarchy this document applies
- [CMP-03 – PCL-AI: the fourteen competencies](#) — the source of the rubric worked in §3.1
- [CAR-07 – Building a portfolio of evidence](#) — the candidate's side of §2
- [CER-07 – Results, scoring, appeals and complaints](#) — what the assessment record has to survive
- [EXB-05 – Standard setting: modified Angoff and the pass mark](#) — the standard-setting method referred to in §4

— Sources and standards

- PCI platform examination configuration, verified August 2026 — 90-minute duration; four-option single-best-answer items; recall, application and analysis cognitive levels, application-weighted; criterion-referenced scoring; configured pass mark 65 %; immediate result with domain breakdown. No item count exists in the platform and none is stated here.
- ISO/IEC 17024, Conformity assessment — general requirements for bodies operating certification of persons. Referenced for its principles — competence defined before assessment, impartiality, separation of training from assessment, documented and reviewable decisions. Described in our own words; no text reproduced. **PCI is not accredited** to ISO/IEC 17024 by any accreditation body.
- All figures in §8 are illustrative and recomputed from first principles.

— Status and version

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